



Shanghai Feiyu Packaging Machinery CO., LTD.

FY-Z240A Packing Machine

Operating Instructions



上海飞羽包装机械有限公司

Shanghai Feiyu Packaging Machinery Co., Ltd



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Outline

- FY-Z240A automatic counting packing machine (hereinafter referred to as packing machine) is a new type of packing machine developed by our company for hardware screws, stamping parts, injection molding and other various regular shapes in the hardware, furniture parts, home appliances, electronics and other industries. The packing machine has accurate counting, high packing efficiency and easy operation and other features. The core control part of this packing machine adopts programmable PLC controller and Delta touch screen. The automation degree and working reliability of the system are at the domestic advanced level. This packing machine implements modular combination. The main modules include packaging machine part, conveying part, counting part, finished product conveying, re-inspection part, scanning code gun input and other auxiliary modules, which can be combined flexibly to meet various needs of customers. Through the preparation interface of the packing machine control system, it can form a fully automatic packing line with other equipment.
- The system adopts Chinese and English touch screen display, and the operation instructions are clear at a glance (such as: length, quantity, speed, etc.)
- Personalized design, can display corresponding company name, phone number and other information according to different manufacturers.
- Application: screws, nuts, washers and other various products with regular shapes.
- Using PLC control system, simple operation, suitable for single item packing.
- Chinese and English menu conversion function, suitable for the international market, and can also be customized according to customer's requirements.
- Optional color code to control the bag length.
- Packing length: 20mm-180mm
- Packaging film drawing adopts stepper motor.
- Packing count: 0-99 million bags, and can be counted cumulatively
- Packaging speed display, real-time display of output per minute.
- Menuized operation interface, and there are operation tips, the operation is extremely simple.
- Intelligent fault diagnosis and display the error location in both Chinese and English (such as: photoelectric error, feeding failure, etc.)
- The equipment has firm sealing, good air-tightness, smooth bag shape, exquisite packaging, high packaging efficiency, sturdiness and durability, and can be equipped with exhaust device, coding machine, labeling machine, finished product conveyor belt, re-inspection device.



Shanghai Feiyu Packaging Machinery CO., LTD.

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Once this product is used, it is deemed that you have carefully read the disclaimer and security matters, understand, approve and accept the full terms and contents of this statement. You undertake to take full responsibility for the use of this product and the possible consequences. You undertake to use our products only for legitimate purposes and agree to any relevant regulations, policies and guidelines under this clause.

Chapter 1 Safety

1.1 Safety precautions

Before installing, operating or maintaining this equipment, please read these instructions carefully and become familiar with this equipment. The following special information may appear in this manual or on the equipment to warn of potential hazards or to remind you of information that clarifies or simplifies a process.

- Distinguish and explain the degree of injury and loss caused by incorrect use of this equipment.

 Danger
This mark indicates related content that is likely to cause death or serious injury

 Notice
This mark indicates related content that is likely to cause injury or property damage

- Corresponding matters are explained with the following graphic marks.


This mark indicates related content that cannot be implemented.


This mark indicates related content that must be implemented.

- Operators must undergo rigorous training (especially safety training) before they can work.
- Operators and equipment maintenance personnel must strictly abide by relevant safety rules.
- This chapter describes all the technical specifications of 240 Packing machine equipment for professional and technical personnel to refer to.
- Related personnel should have basic mechanical and electrical knowledge.
- Disregarding the warning information may lead to dangerous situations, the company is not responsible.



Danger



Do not stack sundries, combustibles, and items unrelated to the equipment around, inside, or on the equipment.

Do not use wet hands for wiring and equipment operation.

Do not modify the equipment.

The equipment safety device and protective cover cannot be removed.

Do not use this equipment in places with water, corrosive, flammable, and explosive places.

Do not turn on the power while servicing the equipment.

After the equipment is repaired, the power cannot be switched on without fully restoring the equipment and the safety shield.

The machine cannot be transported at the same time as the whole machine, and must be disassembled for transport.

No one can stand on the device.

After the sudden power failure is over, when the power supply is restored, there may be a sudden restart phenomenon. Please do not insert objects or sundries into the motion mechanism of the equipment.

Do not wear long skirts or unwrapped long hair to operate equipment. .

Do not adjust the heating temperature of the equipment to more than 150°C.

The equipment should be used in a clean and level place with less dust and no contact with water or oil.

The safety device of the equipment and the function of the safety shield should be checked on time.

The wiring work must be done by an electrical engineer.

Operators of machinery must be trained on site by our company.

The bottom line of the equipment must be reliably grounded.

When an earthquake occurs, the equipment must be reinstalled and debugged to prevent personal accidents.

An emergency stop circuit must be provided for the incoming line of the equipment to ensure that it can be stopped in time and cut off the power supply in an emergency.



An over-current protection device and an over-voltage breaker must be installed at the inlet end to protect the equipment.

Relevant safety confirmations must be made before starting the machine after the earthquake.

The movement of equipment must be broken down into a combined structure.

To maintain the equipment, you must hang the corresponding reminder (no power-on) in the eye-catching position and the main power position.

When performing maintenance on the operating parts of the equipment, the power and air supply must be turned off.

The dressing and dressing of the operator must be concise, and the long hair of the equipment should not be caught.

Before turning on the emergency stop switch and re-energizing, you must pay attention to the surrounding environment (including personnel, equipment and transmission parts) is in a safe state, otherwise it cannot be powered on.

If the person who does not manipulate the switch wants to intervene in the adjustment, debugging or inspection operation of the machine, the person who manipulates the switch must be notified in advance.



Notice



- Do not damage the wire or subject it to excessive external force, gravity, or pinching.
- Do not drop, turn upside down, tilt or dump when transporting.
- Do not subject the equipment to strong shocks.
- The main power supply of the equipment cannot be switched frequently.
- Do not place magnetic objects where there are circuits.
- Do not use equipment to pack fragile products.
- Do not pull external and internal wires at any time.
- Don't try to enter the parameter settings with permission restrictions.
- No modification. Responsible for the safety circuit of the equipment and responsible for the safety of the electrical components of the equipment.
- Do not use hard objects or sharp objects to operate the touch screen.
- It is strictly prohibited to try to enter or force to enter the equipment PLC and touch screen program. Related debugging must be performed by our company's engineers.
- Can't pack hot and melted products.
- The equipment can only package the products required in the purchase contract, and cannot package liquid, hazardous, food-grade products



- During transporting, select the corresponding transport tool according to the quality of machine.
- Observe the specified installation method, installation sequence, and installation direction.
- The air pressure and power supply specifications must meet the nameplate requirements, and the power supply deviation should be used within the range of (+/-10%V, +/-1HZ).
- When the equipment is put into packaging materials, it should be appropriate.
- Maintenance and inspection work should be carried out by professionals.
- Use bare hands to operate the touch screen and buttons.
- Use the specified tools to adjust the mechanical structure to suit different types of materials.
- When the equipment gives an alarm, stop the equipment first before solving the problem.
- The ambient temperature (5-40°C) where the equipment is running.
- The environmental humidity (30-95%) where the equipment is running, without condensation.
- Electrical equipment should be able to work normally at an altitude of less than 1,000m above sea level.
- The equipment should be placed on a level ground with sufficient support and strength.
- If you need to operate an electrical component with power, other electrical components that are not related to the operation must be disconnected from the power supply.
- The equipment needs to be transported or stored, and needs to meet the ambient temperature (-25°C-55°C/70°C).
- When the machine is running, the operator (any one) should press the emergency stop switch (the red button on the conveyor) in time if any abnormal situation is found, and inform other operators.
- The discharge port of the packaging machine must use a conveyor belt device.
- At the end of the equipment installation, the adjusting feet must touch the ground, and it is strictly forbidden to be supported by wheels alone.
- When cleaning the machine and its surrounding area, please do not let water enter the electrical control box and various electrical parts, so as not to damage the components or endanger personal safety.
- A small amount of gas will be emitted when the machine is working, and the installation site should have a good ventilation environment.

1.2 Safe operation instructions

For your personal safety and avoid unnecessary property damage, you must pay attention to the following hazards on the device.

Mark	Explanation	Mark	Explanation
 当心触电 Danger! Electric Shock	To prevent electric shock or misoperation, this operation is only allowed to be operated by professional electrical technicians with relevant qualifications.	 高温 HOT SURFACE	It means that if you don't take corresponding precautions, it may cause personal burns and scalds.
 注意安全 Danger! Safety	It means that if you do not take corresponding careful measures, it will result in major property damage or serious personal injury.	 当心伤手 Danger! Hand Injury	Indicates that if you do not pay attention to the corresponding tips, it may cause serious personal injury.
 当心机械伤人 Danger! Mechanical Wounding	Indicates that serious personal injury may result if corresponding caution measures are not taken.	 当心卷入 Danger! Rolling	Indicates that if you do not pay attention to the corresponding tips, it may cause serious personal injury.
 夹手 SQUEEZE HAND	Indicates that serious personal injury may result if corresponding caution measures are not taken.	 刀片 BLADE HERE	Indicates that there are sharp blades here, which may cause serious personal injury.
 当心皮带伤人 Danger! Leather belt Wounding	Indicates that there is a belt here. If you do not take corresponding precautions, it may cause personal injury to the subspecies.	 接地 Grounding	It means that there must be a reliable grounding, otherwise it may cause electric shock hazard.
MS	Machine Stop state.	MR	Machine Run state.

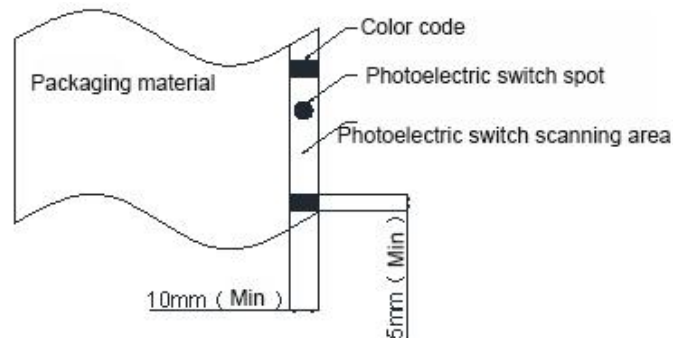


Chapter 2 Main module technical parameters

2.1 Main technical parameters of the packing machine

2.1.1 Packaging materials: PAPER/PE, PET/PE, PP/PE and other composite films;

2.1.1.1 The color code of the packaging material is shown in the figure:



In the photoelectric switch detection area, there must be a certain contrast between the color of the packaging material and the color of the color code, so as not to affect the normal operation of the packaging machine.

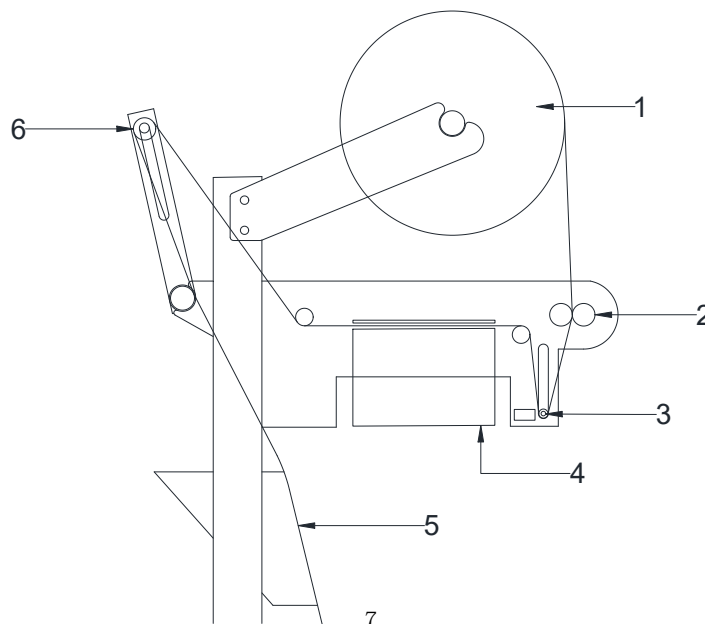
2.1.1.2 The inner diameter of the perforated bone of the packaging material film is 75mm, and the maximum outer diameter of the coil is 300mm;

2.1.1.3 The thickness of the packaging material is $\geq 0.04\text{mm}$, and the material should be smooth, stiff, and composite.

2.1.2 Structure and principle

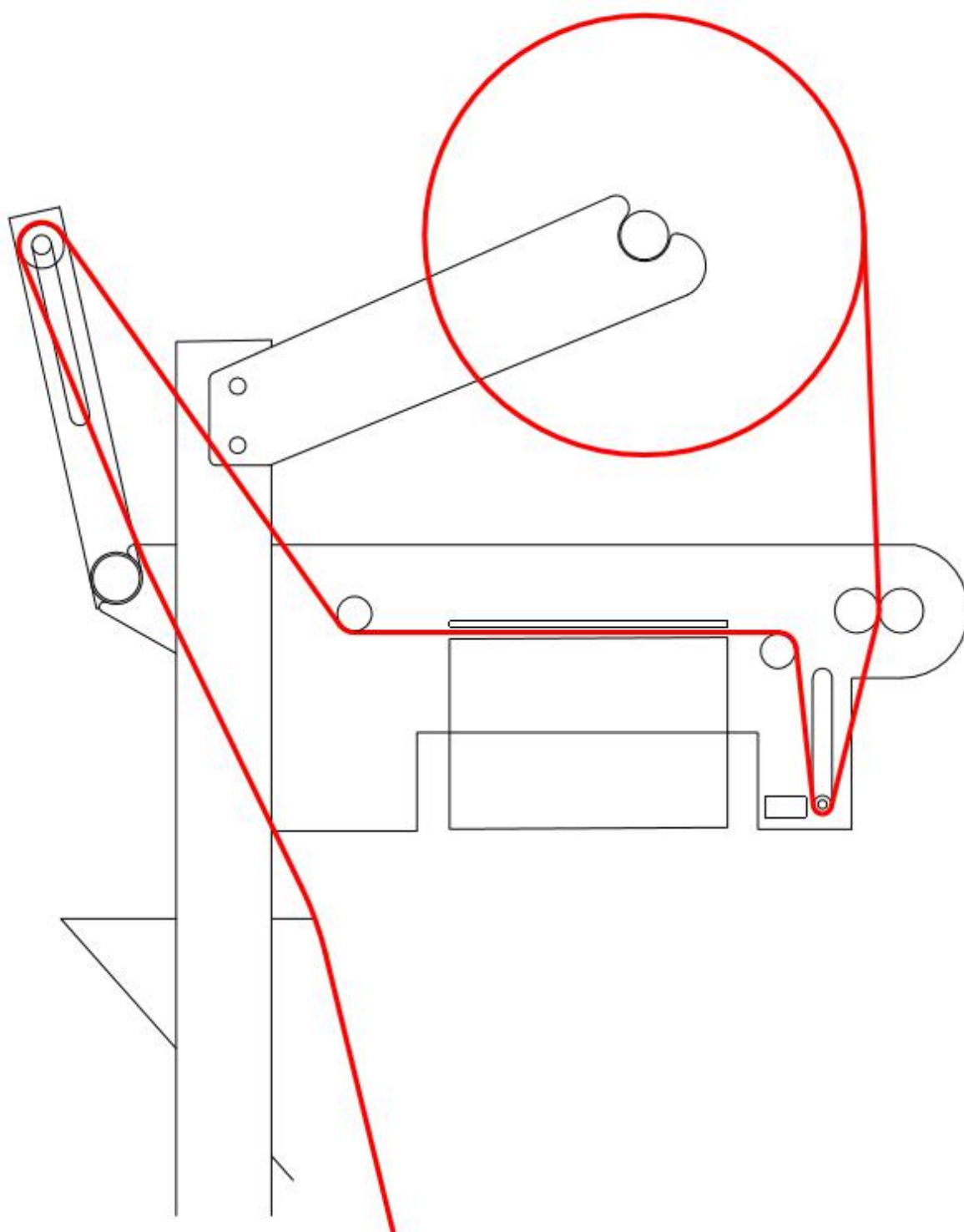
2.1.2.1 Film feeding mechanism:

①Film roll ②Film feeding grinding wheel ③Film feeding induction shaft ④Coding machine ⑤Bag former ⑥Former angle adjustment lever





Membrane route diagram:



The film feeding mechanism is a support and platform that completes the packaging material film storage, film unloading, traction, flattening and forming. The mechanism can also achieve the following functions:

2.1.2.2 Film release control

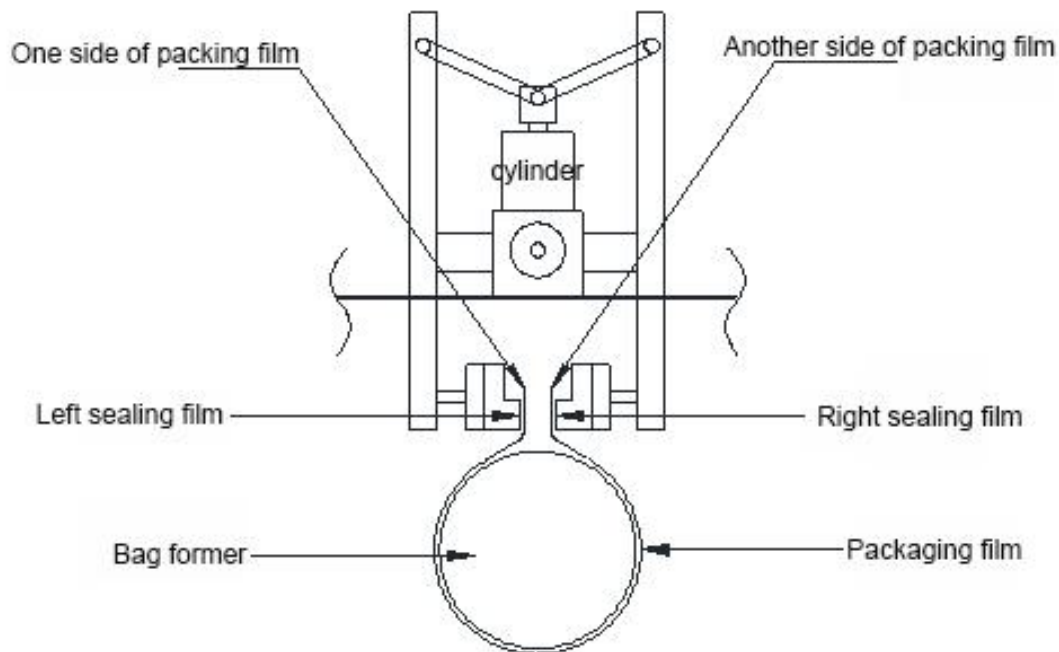
At the beginning of the normal working cycle of the packaging machine, the film pulling mechanism first drives the film on the bracket to move downwards. The tension of the film drives the paper feeding sensor shaft up, the sensor shaft leaves the proximity switch below, the switch indicator light is on, and the film feeding roller motor Work, drive the film to rotate and realize the film feeding action. When the induction shaft is falling, it approaches the proximity switch below, the switch indicator goes off, and the film feeding motor stops, completing a working cycle of film pulling and unloading, and at the same time, it is ready for the next work of pulling and unloading the film.

2.1.2.3 Film exhaustion alarm

In the case of packaging film, every time the film pulling wheel pulls the film, the sensor shaft will inevitably leave the sensor switch once. If there is a film pulling action and the sensor shaft does not leave the sensor switch, the program can judge that there is no packaging film. , After reaching the set time, the machine will alarm to remind the staff to change the film.

2.1.2.4 Vertical sealing system

The longitudinal sealing mechanism mainly completes the heat sealing work on both sides of the packaging film. The packaging film enters the former through the film feeding mechanism, and surrounds the outer wall of the lower barrel. The longitudinal sealing mechanism heats up the overlapping parts on both sides to form a long and simple shape, which is convenient to contain the materials to be packaged. The schematic diagram of the longitudinal sealing mechanism is as shown in the figure:



The vertical sealing action is completed by the vertical sealing cylinder. The left and right sealing molds are clamped by the movement of the cylinder, and the packaging film is sealed by the heating block in the sealing film. In the production process, first select the film adjustment mode on the PLC man-machine interface, start the packaging machine to pack 3-5 empty bags, check whether the left side and the right side of the film shown in the above picture are aligned, and the left side can be 0-3mm more

than the right side. . If there is more on one side, you need to move the whole shaper to which side, and the moving distance is half of the distance on the side of the envelope.

2.1.2.5 Horizontal sealing system

The horizontal sealing mechanism mainly completes the heat sealing work of the top horizontal sealing of the front packaging bag and the bottom horizontal sealing of the rear packaging bag in the continuous production process. In the horizontal sealing process, the two connected packaging bags are cut open by the cutter to complete a horizontal sealing cycle. Before working, adjust the two horizontal sealing blocks to make the teeth fit and the pressure is even.

2.1.2.6 Printing mechanism

The printing mechanism is mainly to realize the thermal printing of the production date and batch number (see the printer manual for details).

2.1.2.7 Control operation part

The electrical control box and operation panel of the packaging machine are installed on one side of the whole rack, which is convenient for the operation and use of the operator. The control panel is mainly composed of touch screen, emergency stop button, power switch, etc. As shown below:

①Touch screen: used to realize the operation control, parameter setting and display related information of the overall packaging unit (see detailed setting instructions for details);

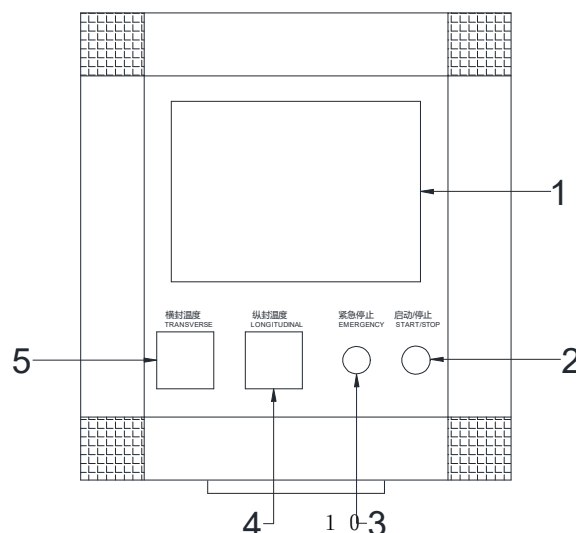
②Start/stop: used to control the start and stop of the packaging machine;

③Emergency stop: The red emergency stop button is mainly used for emergency stop when an emergency occurs during the production process. Press this button and the packaging machine will stop working immediately, and the touch screen will display corresponding information. After pressing the emergency stop button, if you want to restart the equipment, you must first reset the emergency stop switch;

④⑤Temperature controller: control the temperature of each heating plate, and the temperature can be kept stable after 10 minutes after starting up.

2.1.2.8 Pneumatic system

The air circuit system is mainly composed of pneumatic triple parts, solenoid valves, and air cylinders. The functions of the pneumatic triplex are: purify the compressed air entering the air system; adjust the pressure of the air entering the air system; add atomized lubricating oil to the compressed air entering the air system. The solenoid valve controls the action of each cylinder, so that the corresponding actuators of the packaging machine are operated.



2.2 Main technical parameters of vibrating plate counting part

The vibrating plate counting part is mainly composed of a vibrating plate and an optical fiber counting system. First, the vibrating plate transports the workpieces neatly to the fiber counting station. According to different workpiece counting stations, there are generally two modes: track mode and magazine mode. Some special-shaped or larger workpieces will also use the belt counting method.

2.2.1 Track mode

The track mode is usually suitable for workpieces such as screws, nuts, cylinders, etc., which are convenient to enter the track. In this mode, the distance between the ejector pins of the front and rear cylinders into the track is the diameter of a workpiece.

2.2.2 Cartridge mode

The material box mode is suitable for lighter or irregularly shaped workpieces, such as flat washers, spring washers, hole covers, hole plugs, etc. If this kind of workpiece enters the track, it is easy to cause material jam, so it is counted directly at the exit of the vibrating plate. At the same time, in order to increase the speed and increase the storage box, it can be counted while the packaging machine is packaging.

2.2.3 Belt counting mode

Some heterosexual or larger workpieces can neither enter the track nor count well at the exit of the vibration plate, such as pipe fittings and larger injection molded parts, you need to add a belt line at the exit of the vibration plate, and use the high speed of the belt line to pull it apart. The distance between the workpieces, so as to achieve the accurate purpose of counting.

2.3 Main technical parameters of other optional items

2.3.1 Finished product conveyor

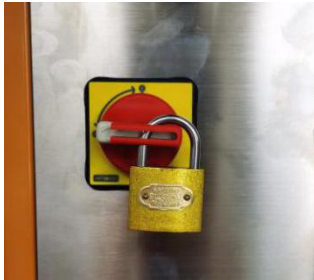
In order to increase the height of the discharge port from the ground and facilitate the receiving of materials, a finished product conveyor can be added to the discharge port of the packaging machine. When using this item, directly insert the aviation plug on the conveyor into the aviation socket under the packaging machine panel. Just one.

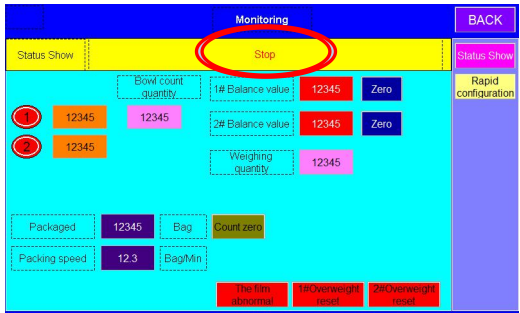
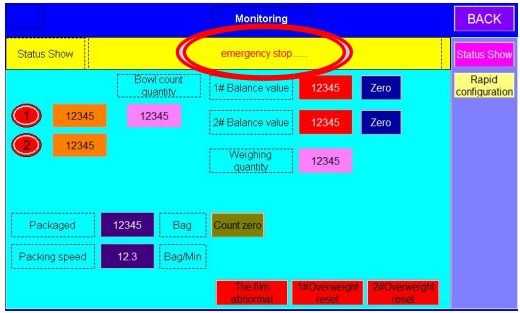
2.3.2 Repetitive weighing

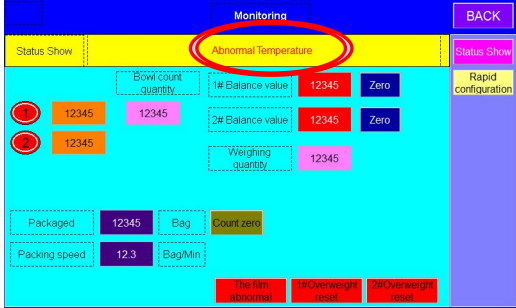
In order to further ensure the accuracy of the quantity of the packaged products, a weighing repeater can be added, and the weight accuracy can reach 0.1g. When in use, directly insert the aviation plug on the recheck scale into the aviation socket under the panel of the packaging machine, and insert the trachea into the tracheal plug at the same position to open the valve.

Chapter 3 Operating instructions

3.1 Normal starting up operation instructions:

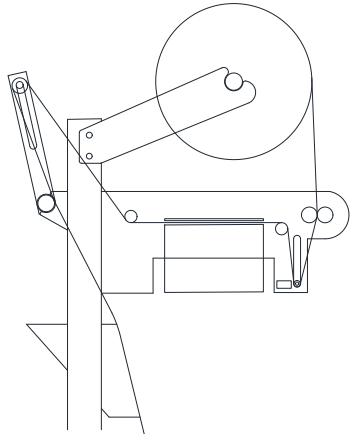
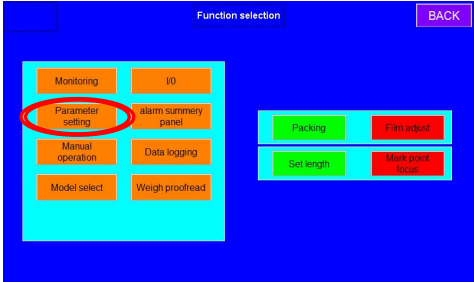
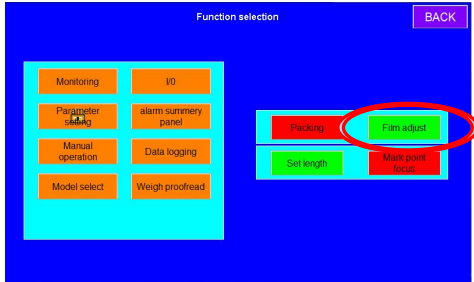
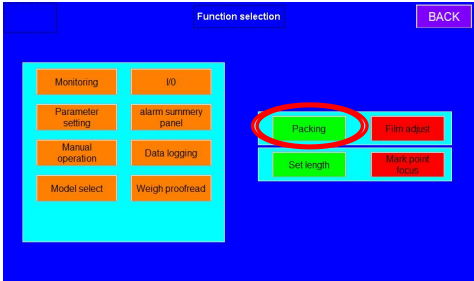
 Normal starting up working rule		
Step	Operation description	Picture shows
1	Preparation and inspection items before use: ① Check the warning signs of the equipment and exclude the relevant staff from maintaining the equipment; ② Check whether the air pressure in the air supply circuit of the equipment is normal; ③ Make sure there is no residue in the scale box. ④ Check that there is no filling bag at the transverse seal. ⑤ Check that there is no clutter in the vibration bowl. ⑥ Make sure the electric box door and mechanical shield are closed.	DANGER!  When the relevant personnel repair the equipment, the power must be turned off and locked to prevent other personnel from opening the switch by mistake and causing personal injury 
2	Turn on the power switch and turn the power rotary switch to "ON" position (usually under the touch screen of the head)	
3	Push on the left air valve switch in the right figure to confirm that the air pressure of the right barometer is between 0.6 MPa and 0.8 MPa.	

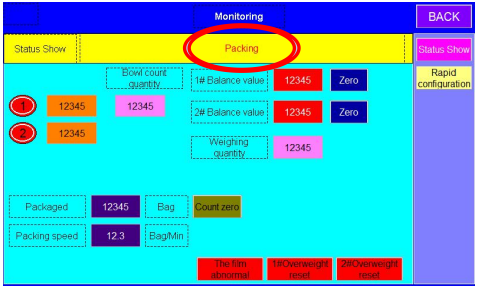
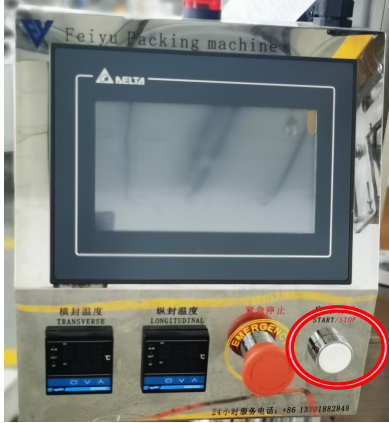
4	Confirm that the power supply of the controller is normal, and the indication of each channel is normal.	
5	Confirm that the power supply of the fiber amplifier is normal, and the light intensity display (red) is 100.	
6	Confirm that the indicator light of air push type recheck scale (optional) is normal.	
7	Confirm the operation screen of human-machine interface - status indication - operation stop.	
8	Press the emergency stop button to confirm the status indication - during emergency stop, confirm whether the emergency stop button is effective.	

9	When the temperature does not reach the set temperature, confirm that the ALM prompt light is on, Screen status indication - abnormal temperature. Confirm that the temperature monitoring of the equipment is normal.	 
10	Start up completed, ready for production.	

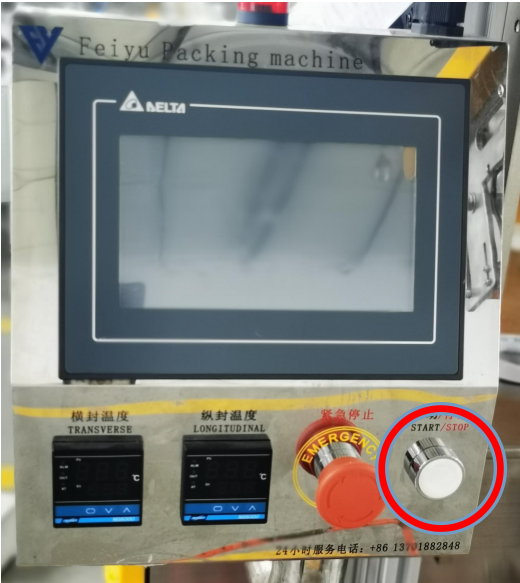
3.2 Normal production operation instructions

 Normal production operating procedures				
Step	Status	Operation description	Necessary tools	Picture shows
1	MS	Prepare the products to be packed, and pour them into the corresponding vibration bowl, adjust the vibration bowl and track required by the products to adapt to the current items	Hex wrench Allen wrench	

2	MS	Put on the packaging film that needs to be used	4mm Allen key scissors	
3	MR	Enter the operation interface, select the corresponding authority according to their own operation authority, and the general staff select - operator		
4	MR	Before the formal production, the relevant management personnel should set the production related parameters (see the detailed setting instructions for details)		
5	MR	Select the film adjustment option, test whether the packaging head can run normally after the temperature reaches, and adjust the length of the bag according to the size of the item		
6	MR	Switch to production option and press start / stop key to start production		

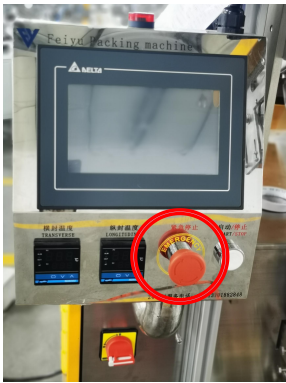
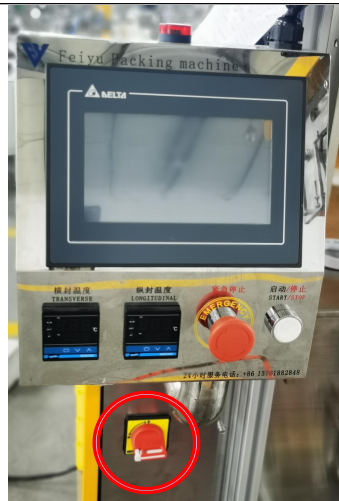
7	MR	Observe the normal operation of each component of the packaging machine, and check whether the quantity and speed of the finished products meet the requirements, and make corresponding adjustments.		
8	MR	After the debugging is OK, continuous production can be carried out, and the start / stop button can be pressed when the machine is stopped		

3.3 Normal shutdown operation instructions

 Normal turn off operating procedures		
Step	Operation description	Picture shows
1	When preparing to stop the machine, press the start / stop button. After the last box of materials enters the bag, the machine stops running and checks whether the machine is abnormal	

2	After checking that the machine is normal, turn off the power by turning the power switch	
3	Push the gas valve switch in the opposite direction to turn off the air source, and then turn off the work	

3.4 Emergency operation instructions

 Danger Emergency operation procedures		
Emergency:	When the following conditions occur during the operation of the equipment, please implement the following emergency operation procedures: 1. There is violent explosion sound in the control cabinet; 2. Fire around; 3. power supply cable on fire or see obvious overload burning smell; 4. Chain box is stuck or stuck in people or objects; 5. People or objects are clamped in the seal; 6. Other situations endangering personal and property safety.	
Step	Operation Description	Picture shows
1	Press the emergency stop button on the control cabinet	
2	Cut off the main power switch on the control cabinet	
3	In case of injury, take the injured to the medical room or the nearest hospital for treatment	
4	In case of equipment fire and smoke, take appropriate fire extinguisher nearby to extinguish the fire	
5	Keep the scene and call the relevant equipment maintenance personnel	

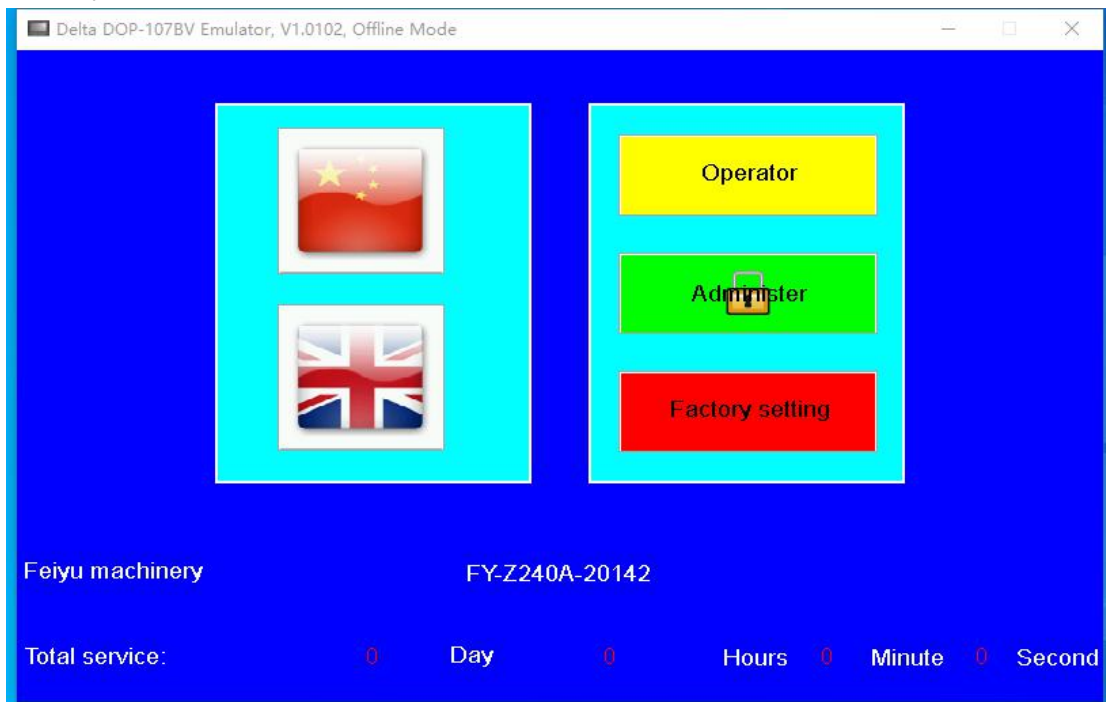
Chapter 4 Detailed setting instructions

4.1 Main Interface

After the power is connected, enter the boot interface:



Click any position to enter the main interface:

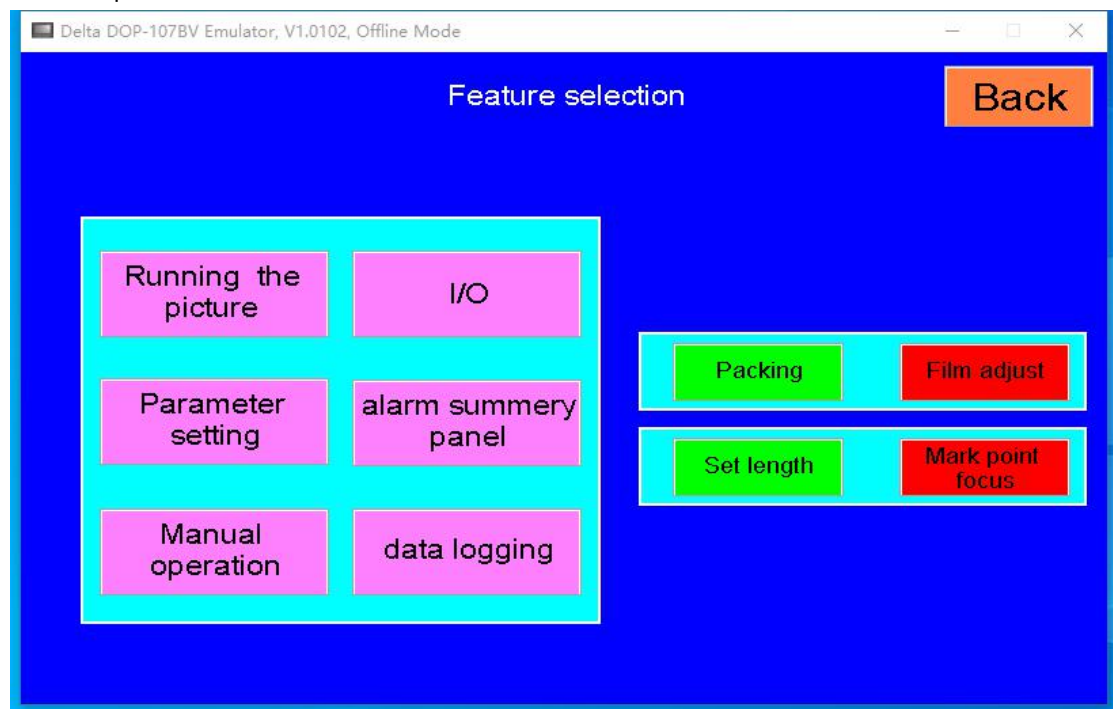


4.1.1 Language selection: click the corresponding icon to select the language and switch between Chinese, English and Spanish languages;

4.1.2 Permission selection: the operator's permission can be used in normal production. If the parameter setting needs to be changed, it needs to be obtained permission of Administrator(Password: 1357) or Factory setting(65535)

4.2 Function selection

Click the operator to enter the function selection interface:



In this screen, the user can select the packaging function according to the actual situation, and enter eight lower level menus, including monitoring screen, I / O monitoring, parameter setting, alarm screen, manual screen, data record, mode selection and scale calibration screen.

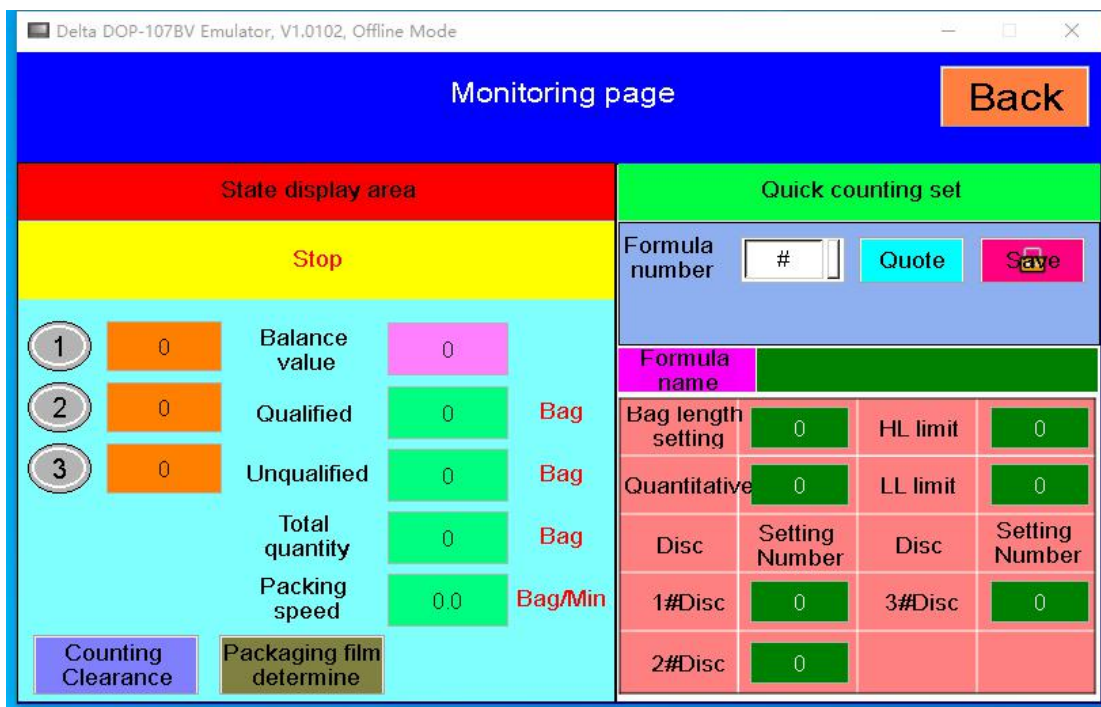
Introduction of packaging function button:

4.2.1 Production / film adjustment: when selecting production, the machine will enter into normal production state; when selecting film adjustment, only the packaging head will run. This mode can be used for the adjustment of packaging film installation, replacement and film sticking.

4.2.2 Fixed length / calibration: when selecting the fixed length, the production will be carried out according to the set length of the packaging film; when selecting the calibration, the production will be carried out according to the position induction of the color standard of the packaging film.

4.3 Monitoring

Click on the monitoring screen to enter:



4.3.1 Status display: the main function of this screen is to monitor the real-time status of the machine during normal production

4.3.1.1 Status indication: display the current alarm status, and at the same time, the text behind the status indication shows the current alarm. Related status indication and solution:

Instructions	Meaning	Solution
Abnormal temperature	The temperature is too low or too high	1. When starting the machine, wait until the temperature reaches the set temperature, and the prompt disappears before starting the machine; 2. The prompt appears during operation: it is necessary to suspend the operation of the equipment and wait for the temperature to rise. If the set temperature is still not reached after 5 minutes, please check the heating circuit; 3. Check whether the heating element is leaking electricity and causing excessive temperature.
Emergency stop	The emergency stop button was pressed	Confirm the intention of pressing the button and release the emergency stop button after confirming the safety of the equipment.
Stop running	Equipment access standby status	It means that the equipment has no system problem, and the machine can be started up after detecting the surrounding materials
Running	Running	Keep away from the dangerous area. The equipment may move at any time

* material supplement alarm (Note: * represents the station number of the vibrating bowl)	The station counts abnormally	1. The station is short of materials, please add materials 2. The material in the station is stuck. Please adjust the equipment. 3. The speed of the material passing through the optical fiber is fast, so it can't be detected, and false alarm will be given. You can observe whether the material passes normally through the current count in the 4 area of the screen. If there is normal counting, it can be ignored.
No packaging film	Packaging film abnormal	1. Please replace the packaging film after use; 2. The packaging film falls off from the film pulling wheel and needs to be re clamped; 3. There is no need to pay attention to the long packing film at the sensing part of the packaging film.
Quantitative complete	The output reaches automatic shutdown	If you need to restart, issue a new formula and start it up.
No. * scale is overweight (Note: * "stands for scale number)	The scale is overweight	Click the No. * overweight button under the monitoring screen to continue weighing and packaging.

4.3.1.2 1#、2# bowl: Display the real-time blanking quantity of the corresponding feeding tray

4.3.1.3 Supplement quantity of bowl: Display the quantity of material to be fed by the feeding bowl, which is obtained by subtracting the number of packages set by the scale.

4.3.1.4 Scale value: Display the real-time weighing reading of the corresponding scale.

4.3.1.5 Number of scales completed: Displays the number of counts completed by the scale.

4.3.1.6 Packed: Used to display the total number of packages completed.

4.3.1.7 Packing speed: Used to display the real-time speed (packets / min) of the current package.

4.3.1.8 Count clear: Clear statistics of generated bags.

4.3.1.9 No film alarm: When the film feeding mechanism is abnormal, the prompt will flash here. Click to reset after troubleshooting.

4.3.1.10 1#、2# overweight: When the weighing quantity of the corresponding weighing mechanism is greater than the set quantity, the machine will alarm and remind. This button can reset the alarm and continue weighing.

4.3.2 Quick setting

Monitoring						BACK
Formula No	1234	ABCDEFGHIJKLMN		Quote	Save	Status Show
Formula name	ABCDEFGHIJKLMN					Rapid configuration
Packaged	1234	1# Quick complement	12345	1# High-speed	12.3	
1# Collecting Time	12.3	1# Mid complement	12345	1# Mid-speed	12.3	
2# Collecting Time	12.3	1# Slow complement	12345	1# Low-speed	12.3	
Quantitative packaged	12345	2# Quick complement	12345	2# High-speed	12.3	
Bag length set	1234	2# Mid complement	12345	2# Mid-speed	12.3	
1# Set the number	1234	2# Slow complement	12345	2# Low-speed	12.3	
2# Set the number	1234	1# balance?scale	2# balance?scale	1# Bowl	2# Bowl	

4.3.2.1 Save formula group: after parameter setting, input the saved formula number and formula name, and click Save button to save the current formula group. When using the next time, select the corresponding formula number and directly click call to quickly complete the data setting, which is convenient and efficient.

4.3.2.2 Number of packages: set the number of packages per package.

4.3.2.3 Storage time: in order to improve the packaging speed, a storage valve will be installed on the top of the scale box. When the materials enter the weighing box, the storage valve can save materials in advance. When the materials in the scale box are completely closed after pouring out the materials, the storage valve will open to pour the materials into the scale box. This time is the time for the storage valve to save materials.

4.3.2.4 Fixed number of packages: after setting, the packing machine stops working when the packed quantity reaches the set quantity.

4.3.2.5 Bag length setting: set the bag length of the packaging bag. This setting is only valid for the fixed length packaging mode.

4.3.2.6 Set number of bowl: set the blanking quantity of the feeding bowl. When the weighing switch is on, the system will automatically calculate the quantity of the vibrating bowl according to the weight of the weighing box. At this time, this setting is invalid. This setting is only valid when the vibrating bowl is opened separately. If it is necessary to adjust the bowl separately, this setting can be used.

Notes for adjusting bowl:

- ① The vibration bowl should adjust a reasonable speed and the screw should run forward normally;
- ② The last row of material should be controlled in the part of drawing and inserting board;
- ③ The first blowing position is to blow down and accelerate the standing screw to increase the amount of hanging material;
- ④ The hanging coaming should be adjusted outward and downward to ensure that the standing screw can not enter. If it enters, there will be jamming.

4.3.2.7 Fast mantissa: control the number of magnetic and belt elevator working with fast speed, when the scale box detects

When the quantity is less than the number of packages minus the fast mantissa, the hoist works at a fast speed.

4.3.2.8 Medium-speed mantissa: Control the number of magnetic belt elevators working at medium speed. When the scale box detects that the quantity is greater than or equal to the number of packages minus the fast mantissa, but is less than the number of packages minus the medium-speed mantissa, the magnetic belt The hoist works at medium speed.

4.3.2.9 Slow mantissa: control the number of magnetic belt hoists working at slow speed. When the weighing box detects that the quantity is greater than or equal to the number of packages minus the middle speed mantissa, but less than the number of packages minus the slow mantissa, the magnetic force The belt elevator works at a slow speed until the scale box detects that the quantity is greater than or equal to the number of packages minus the slow mantissa, the elevator stops working.

Note: The number of slow mantissa, medium speed mantissa, fast mantissa, and the number of packages must be set as increments, and the subsequent quantity cannot be less than the previous quantity.

4.3.2.10 Fast speed: Set the fastest working speed of the magnetic belt hoist. When the detection quantity of the scale box is before the fast mantissa, the hoist will always work at this speed.

4.3.2.11 Medium speed speed: Set the medium speed working speed of the magnetic belt hoist. When the detected quantity of the scale box is between the fast mantissa and the medium speed mantissa, the hoist will always work at this speed.

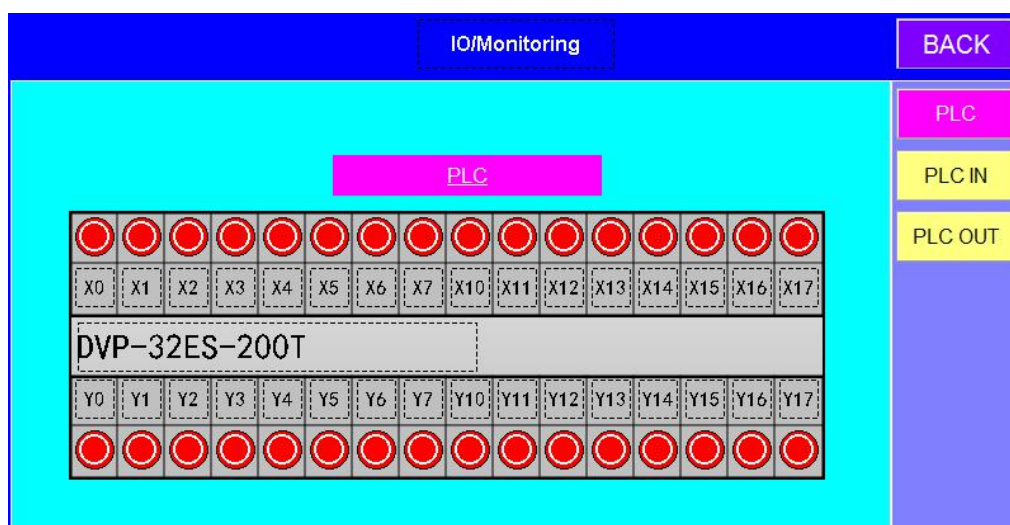
4.3.2.12 Slow speed: Set the slow working speed of the magnetic belt hoist. When the detected quantity of the scale box is between the middle speed mantissa and the slow mantissa, the hoist always works at this speed.

Note: The setting value of fast speed, middle speed and slow speed must be decreasing.

4.3.2.13 Scale switch and pan switch: You can choose whether the scale or pan is required to participate in the work, red off and green on.

4.4 I/O monitoring

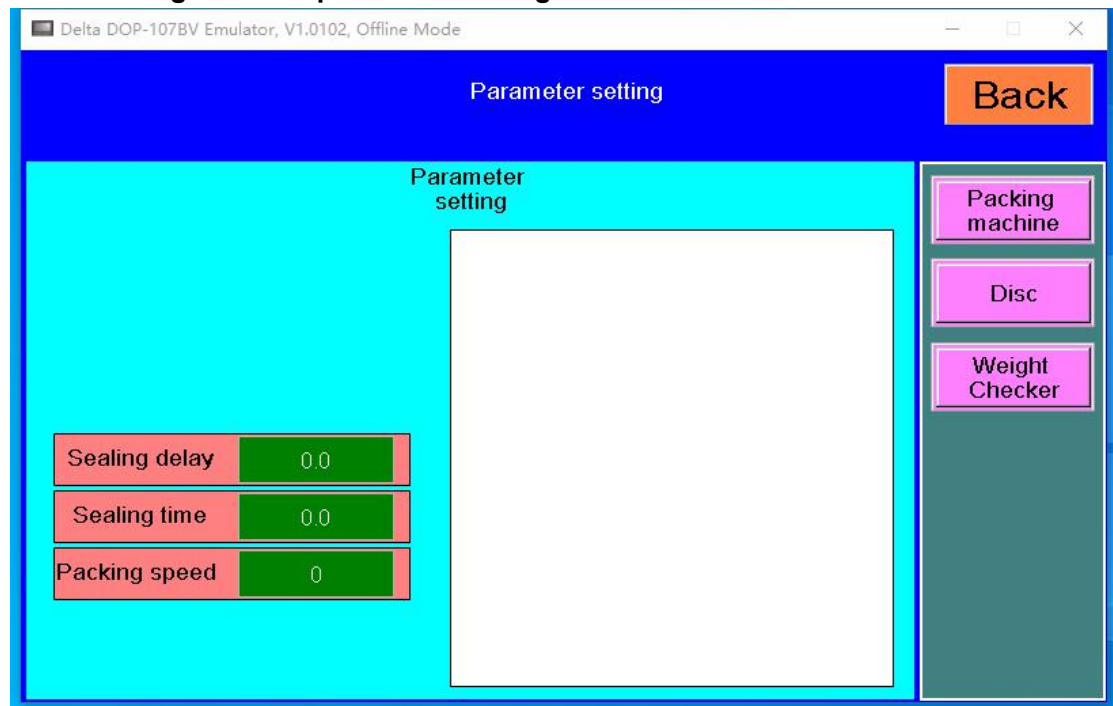
Monitor the PLC running status, usually used to detect and eliminate abnormalities or faults.



4.5 Parameter setting

Entering this screen requires administrator authority, password: 1357; or entering the factory setting interface, the factory setting interface contains all the settings within the administrator authority, access method: click on the factory setting icon on the main interface, and long press the four words of factory settings 3-5 After seconds, enter the password 65535 and you can enter. This section is explained by the factory setting authority.

4.5.1 Packing machine parameter setting



4.5.1.1 Laminating delay time: refers to the time to start the laminating after the material box is poured. Because some materials are lighter, the delay time for activation to prevent them from falling to the bottom of the bag in a short time can be set according to requirements.

4.5.1.2 Film combination time: refers to the film sealing time of the film sealing system. It is recommended to generally set to 0.7S and fine-tune according to the actual situation.

4.5.1.3 Packaging speed limit: limit the packaging speed of the packaging machine. For example, when using a re-inspection scale, due to the limited detection speed of the re-inspection scale, in order to prevent the packaging speed from being too fast, the packaging speed needs to be limited under certain circumstances.

4.5.1.4 Conveyor stop delay: refers to the running time of the finished product conveyor.

4.5.1.5 Quantitative stop: After setting this item, the machine will stop automatically when the packaged quantity reaches this set value.

4.5.1.6 No film alarm: refers to the time when the system fails to detect the packaging material.

4.5.1.7 Delay of pushing plate: in order to prevent the delay time of the start of the pushing plate before the packaging bag has completely fallen to the bottom.

4.5.1.8 Pushing time: refers to the time for pushing the material.

4.5.1.9 Membrane opening time: refers to the time when the horizontal seal is fully opened, and the film system can only be pulled after the horizontal seal is fully opened

Continue to pull down the membrane.

4.5.1.10 Cutter delay: refers to the delay time before the cutter starts. The cutter should start after the film is clamped, so this time setting must be greater than the film delay time.

4.5.1.11 Cutter time: refers to the working time of the cutter. The cutter should be retracted before the film is opened. Therefore, the cutter time plus the cutter delay time should be less than the total time of filming.

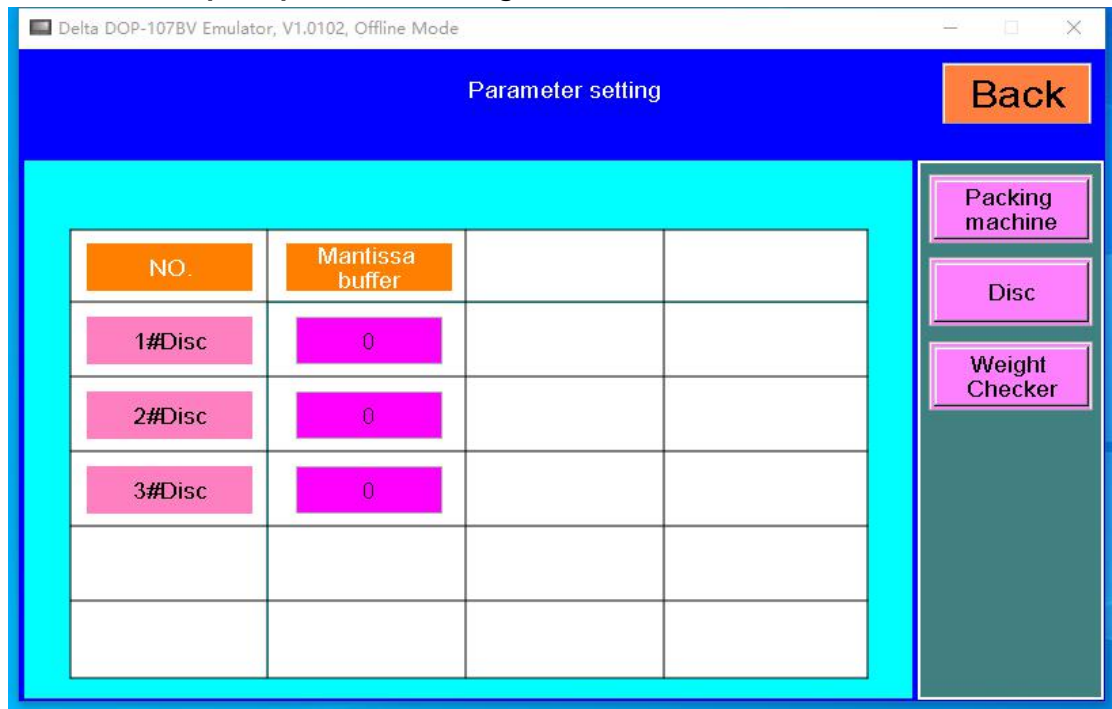
4.5.1.12 Fine adjustment of bag length: Since the bag length is controlled by the film pulling wheel, the actual bag length may deviate from the set value. This setting can adjust the ratio between the set value and the actual bag length to make the actual bag length It is equal to the set value.

4.5.1.13 Film pulling speed: refers to the speed of film pulling motor.

4.5.1.14 Film pulling delay: It refers to the time from the film feeding sensor axis to the film pulling.

4.5.1.15 Acceleration and deceleration time: refers to the slow speed time when the film pulling system is running or stopping, to prevent the packaging film from being damaged due to excessive speed changes.

4.5.2 Vibration plate parameter setting



NO.	Mantissa buffer		
1#Disc	0		
2#Disc	0		
3#Disc	0		

Buttons on the right: Back, Packing machine, Disc, Weight Checker.

4.5.2.1 Number of buffers: This setting is only for models in orbital mode (as shown in the figure below), because the workpieces accumulated in the orbit are flushed out at high speed when starting to discharge. If there is no buffer number, when the optical fiber senses When the number is set, the cylinder may not have time to intercept, which may cause multiple packets. When the number of buffers is set, the optical fiber senses that the number of buffers is subtracted from the number of buffers (for example, the number of settings is 50, The number of buffers is 8, that is, when the optical fiber senses 42), the front and rear cylinders installed at the discharge port will intervene in the work to discharge the remaining materials one by one to ensure the accuracy of the counting.



4.5.2.2 Alarm time: refers to the time from when the vibration plate starts to be abnormal to the alarm.

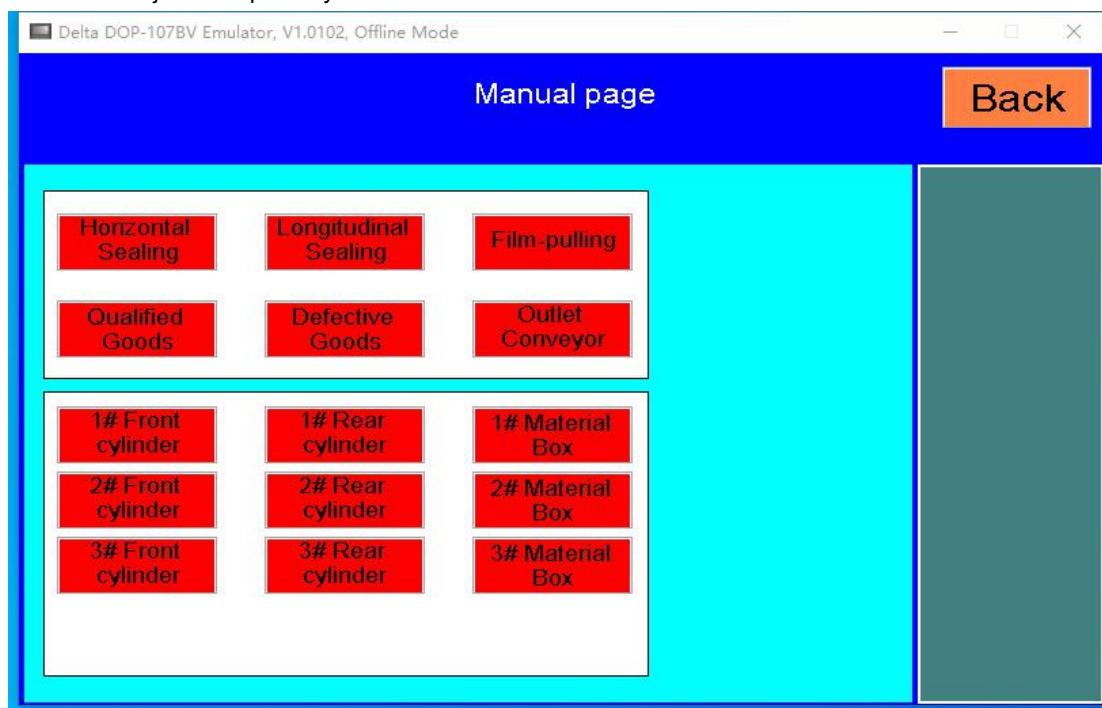
4.6 Alarm screen

This interface will record the abnormal alarm records of the machine, including trigger time, recovery time, and alarm messages, which are convenient for statistical viewing.

Alarming			BACK
Trigger	Recovery	Message	
hh:mm:ss mm/dd	hh:mm:ss mm/dd	####	
hh:mm:ss mm/dd	hh:mm:ss mm/dd	####	
hh:mm:ss mm/dd	hh:mm:ss mm/dd	####	
hh:mm:ss mm/dd	hh:mm:ss mm/dd	####	
hh:mm:ss mm/dd	hh:mm:ss mm/dd	####	
hh:mm:ss mm/dd	hh:mm:ss mm/dd	####	
hh:mm:ss mm/dd	hh:mm:ss mm/dd	####	
hh:mm:ss mm/dd	hh:mm:ss mm/dd	####	
hh:mm:ss mm/dd	hh:mm:ss mm/dd	####	
hh:mm:ss mm/dd	hh:mm:ss mm/dd	####	

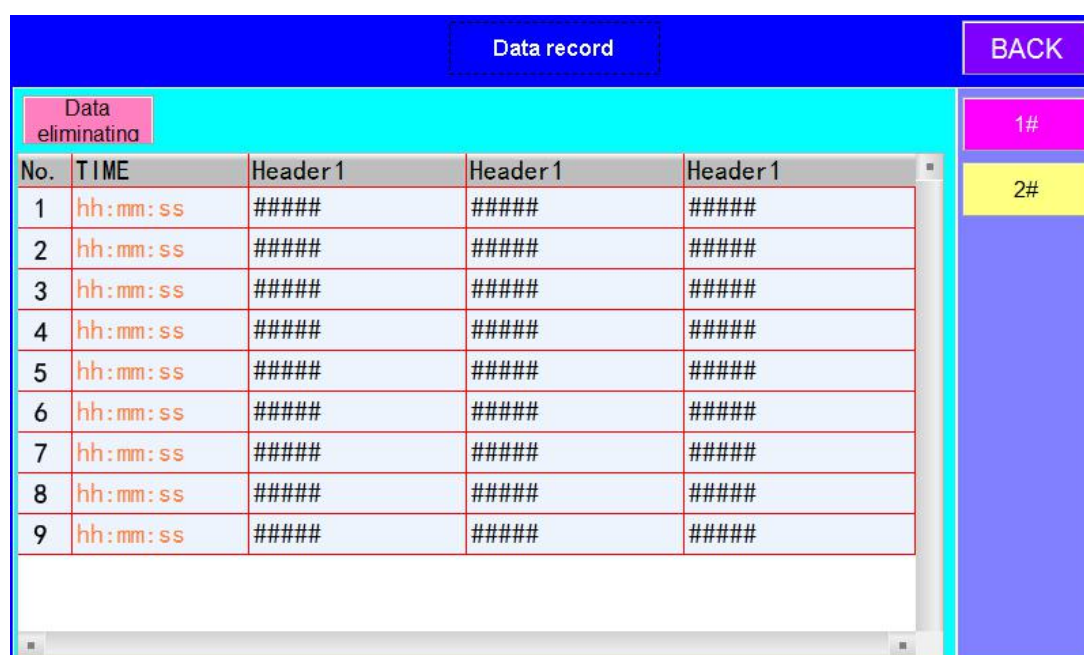
4.7 Manual screen

This interface can manually open and close all the component functions of the packaging machine, and can be used to detect whether each component can run normally or manually operate components that need to be adjusted separately.

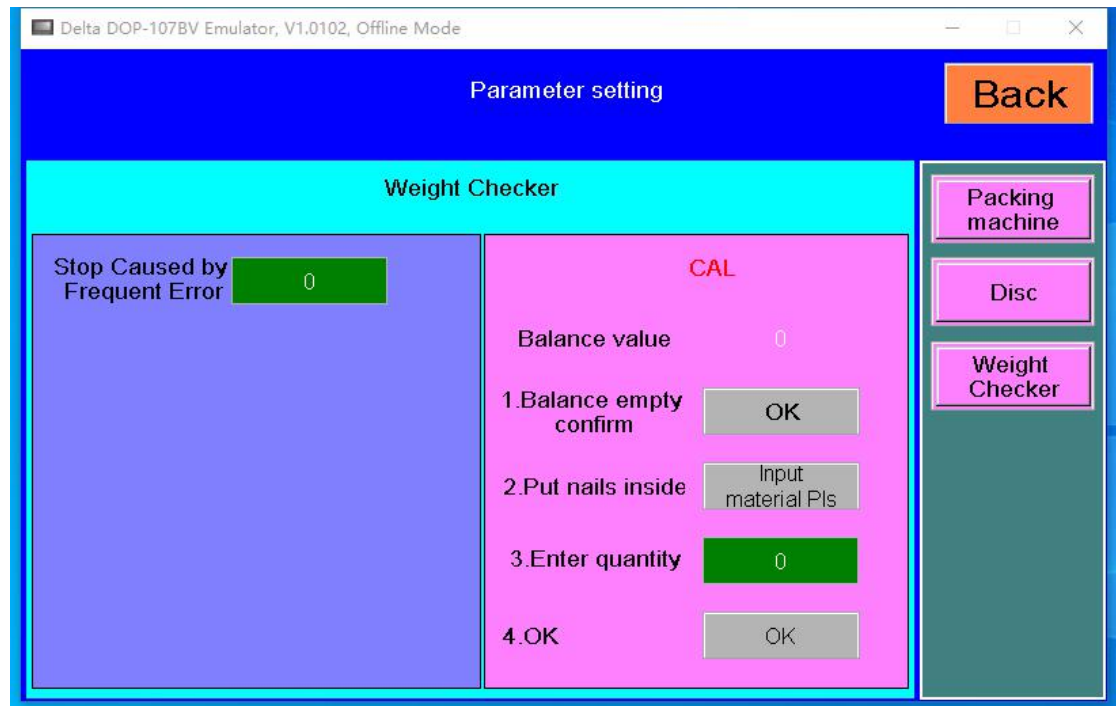


4.8 Data recording

This interface is aimed at models with recheck scales. It can record the work records of recheck scales. Click Data Clear to clear the statistical data.



4.9 Calibration screen



Calibration process:

- ① Clean up the scale box and confirm that there is no debris, then click to confirm the empty scale;
- ② After the scale reading is reset to zero, put the material on the recheck scale and click Please put the material;
- ③ After entering the number of materials, enter the number of materials to be put in (Note: In order to ensure the accuracy of weighing, the number of materials put in should be as close as possible to the number that needs to be packed);
- ④ After entering the number, click OK to calibrate the scale successfully.

Chapter 5 Common problems and solutions

Common problems and solutions:

Common faults	Inspection part	Treatment method
The temperature does not rise or fall	1. Check whether the heating plate is damaged; 2. Check whether the heating circuit is open.	1. Replace; 2. Restore or replace.
The temperature exceeds the set value and continues to rise	1. Check whether the heating solid state relay is always on 2. Check whether the heating element is leaking.	1. Replace; 2. Replace.
Cutting the bag continuously	1. Check whether the blade is worn or not; 2. Check whether the sealing film is misplaced; 3. Check whether the sealing film opens and closes smoothly; 4. Check whether the air pressure meets the needs; 5. Incorrect parameters at the moment of cutting.	1. Replace; 2. Adjust the sealing film to the alignment; 3. Regular lubrication; 4. Check the gas source; 5. Readjust the cutter parameters.
Inaccurate or non-counting	1. Check whether the light display value of the optical fiber sensor is 100_; 2. Check whether the fiber optic probe is dirty; 3. Whether the optical fiber is broken.	1. Adjust the total brightness after cleaning the dirt (refer to the p_t setting in the manual); 2. Clean dirt and adjust brightness; 3. Replace.
Cutting knife, sealing film often cut, clip materials	1. Test whether the bag length can reach the required capacity; 2. Check whether the bag length is uniform 3. Whether the blanking time and the sealing time match.	1. Adjust the bag length; 2. Refer to [Uneven bag length] item; 3. Adjust the sealing time parameters to the packaging machine parameters.
The packaging bag sticks to the seal and causes accumulation	1. (Main problem) The packaging film material has two layers attached together. The inner layer is sticky after being heated, and the outer layer is not sticky. Then when the packaging film is When the back seal is not aligned, it will cause the hot melt material of the inner layer of the film to remain on the seal and accumulate for a long time. It is easy to produce sticky bags and stick to the bags. 2. The glue left on the seal has not been cleaned for a long time. 3. The contents of the bag are too light to fall freely. 4. Long-term use of the sealed iron block may often press on the material, resulting in burrs on the sealed iron piece and easy to hang the bag.	1. Adjust the position of the former. If it is impossible to align the film completely, the hot melt material in the inner layer must also be turned over, pointing to the belt surface and facing away from the sealing iron block; 2. Buy a steel wire brush with a certain degree of rigidity and brush it frequently; 3. Consider fixing a thimble on the chassis panel at the lower part of the seal, so that when the seal is opened, the bag remaining on it just hits the added thimble; 4. Repair or replace the seal.
Uneven bag length	1. Check the resistance on the film threading route; 2. Check whether the film pulling wheel is clamped; 3. Check whether there is rotation resistance of the film pulling wheel; 4. Check whether the pulling speed is too fast.	1. Adjust the piercing route, center; 2. Adjust the mechanical structure or replace the tension spring; 3. Adjust the mechanical structure and regular lubrication; 4. Adjust the parameters.

Chapter 6 Maintenance instructions

6.1 Maintenance and maintenance instructions

In the process of operating the equipment, the operator should pay close attention to the operation of the equipment, make a record of the equipment operation, and record the operation of the equipment in the class in detail for the equipment maintenance personnel to refer to when inspecting and maintaining the equipment.

If an abnormality is found, the machine must be shut down immediately, and the maintenance personnel must be notified to deal with it, and the machine can be restarted after troubleshooting.

Equipment maintenance personnel shall conduct regular inspection, maintenance and maintenance of the equipment.

Before servicing the equipment, you must disconnect the power supply and follow the relevant safety regulations. Any fixed protection device needs to be opened or removed due to maintenance, and must be restored to its original position intact after maintenance.

6.2 Daily maintenance

Before turning on the power every day, you must check whether there are debris on the conveyor belt, such as wrenches, screwdrivers, etc.

Observe whether the power supply and air supply are normal, and whether the power cord and air pipe are damaged.

Before leaving get off work every day, turn off the power and air supply, and do the following:

To clean the machine, you can wipe the outer surface of the machine with a dry soft cloth. If there is too much dust and cannot be wiped, you can use compressed air to blow it away.

If there is sticky material on the conveyor belt, wipe it with a wringed damp cloth.

When wiping, pay attention to observe whether there are any loose screws on the machine. If there are any loose screws, they should be tightened immediately.

6.3 Daily maintenance

Maintenance items	Frequency	Maintenance method	Tools	Picture
Horizontal seal lubrication	Weekly	Grease the sliding rod directly	brush	 A photograph of a machine's horizontal seal assembly. Two vertical sliding rods are circled in red, indicating the points for direct greasing.
Pull film gear lubrication	Weekly	Grease the gear directly	brush	 A close-up photograph of a pull film gear mechanism. The gear teeth are circled in red, showing the location for direct greasing.
Vibration plate counting fiber head	Every 2 days	Ash removal	Cotton swabs and industrial alcohol	 A photograph of a vibration plate counting fiber head. A red circle highlights the fiber head area, indicating where ash removal should be performed.
Horizontal seal	Every day	Remove sticky material	Wire brush	 A photograph of a machine's horizontal seal assembly. A red circle highlights the seal area, indicating where sticky material should be removed with a wire brush.

Daily maintenance instructions:

The types of grease to be applied are generally: Extreme pressure grease (model NLGI2) or Calcium-based grease

The industrial alcohol used for cleaning is generally: the content of industrial ethanol is 95% (C₂H₅OH)

6.4 Common maintenance of electrical components

Carry out cleaning and dust removal work once a week, and dust and clean the electrical components in the electrical control cabinet;

It is best to stabilize the terminals of the machine every other week after the machine continues to work to prevent poor contact;

Pay attention to check whether the motor wire is peeling off, if so, please replace the wire immediately to prevent short circuit and electric shock.